

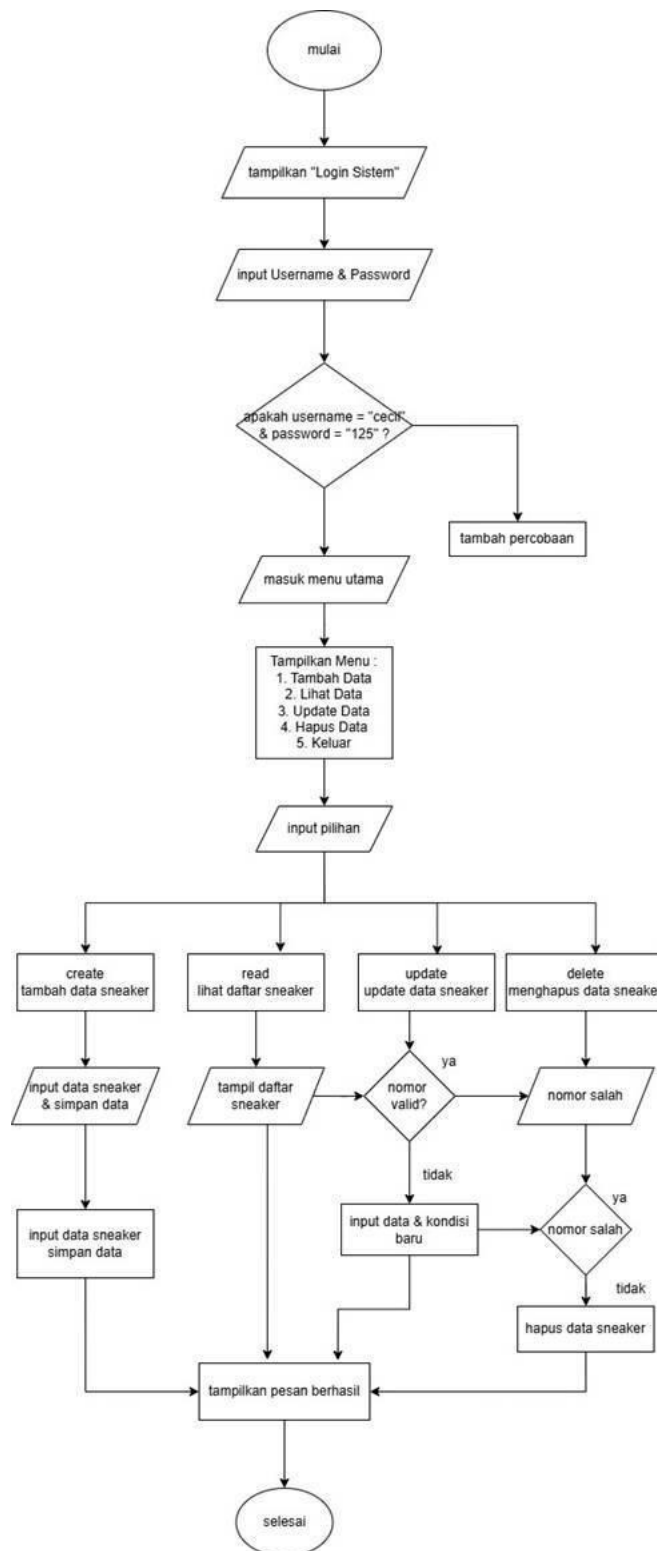
**LAPORAN PRAKTIKUM**  
**POSTTEST 6**  
**ALGORITMA PEMROGRAMAN LANJUT**



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**2026**

## 1. Flowchart



Gambar1.1 Flowchart

## 2. Deskripsi Singkat Program

Program ini digunakan untuk mengelola data koleksi sneaker menggunakan konsep CRUD (Create, Read, Update, Delete). Program ini dirancang untuk menyimpan dan mengelola informasi sneaker seperti brand, model, warna, ukuran, harga, dan kondisi sepatu. Untuk mengelola data sneaker yang dilengkapi dengan fitur **login** dan **sorting (pengurutan data)**.

### Login User

- Pengguna harus memasukkan username dan password
- Maksimal 3 kali percobaan

### Menampilkan Data

- Menampilkan seluruh data sneaker yang tersedia

|                |             |                |               |
|----------------|-------------|----------------|---------------|
| <b>Sorting</b> | <b>Data</b> | <b>(Sesuai</b> | <b>Modul)</b> |
|----------------|-------------|----------------|---------------|

Program menerapkan 3 metode sorting, yaitu:

- **Bubble Sort (Ascending)**

Mengurutkan data berdasarkan brand (A–Z)

- **Selection Sort (Descending)**

Mengurutkan data berdasarkan harga (tertinggi ke terendah)

- **Insertion Sort (Ascending)**

Mengurutkan data berdasarkan ukuran (kecil ke besar)

Program ini merupakan pengembangan dengan menambahkan fitur **searching** pada data sneaker. Terdapat dua metode pencarian yang digunakan, yaitu **Linear Search** untuk mencari berdasarkan brand (string) dan **Binary Search** untuk mencari berdasarkan harga (integer).

Sebelum menggunakan Binary Search, data diurutkan terlebih dahulu karena metode ini membutuhkan data terurut. Program akan menampilkan hasil pencarian jika data ditemukan atau pesan jika data tidak tersedia.

### 3. Source Code

```
#include <iostream>
using namespace std;

struct DetailSneaker{
    string warna;
    int ukuran;
};

struct Sneaker{
    string brand;
    string model;
    int harga;
    string kondisi;
    DetailSneaker detail;
};

struct User{
    string username;
    string password;
};

Sneaker dataSneaker[100];
int jumlahData = 0;
User akun = {"cecil", "125"};

bool login(User user){
    string username, password;
    int percobaan = 0;

    while(percobaan < 3){
        cout << "\n=== LOGIN ===\n";
        cout << "Username : "; cin >> username;
        cout << "Password : "; cin >> password;

        if(username == user.username && password == user.password){
            cout << "Login berhasil!\n";
            return true;
        } else {
            cout << "Login gagal!\n";
            percobaan++;
        }
    }
}
```

```

    }
}
return false;
}

void isiDataAwal(Sneaker *data, int &jumlah){
    data[0] = {"Nike","AirForce1",1500000,"Baru",{"White",42}};
    data[1] = {"Adidas","Yeezy350",4200000,"Baru",{"Black",43}};
    data[2] = {"Converse","ChuckTaylor",850000,"Bekas",{"Red",41}};
    jumlah = 3;
}

void tampilData(Sneaker *data, int jumlah){
    cout << "\n=== Data Sneaker ===\n";
    for(int i=0; i<jumlah; i++){
        cout << i+1 << ". "
            << data[i].brand << " "
            << data[i].model << " | "
            << data[i].detail.warna << " "
            << data[i].detail.ukuran << " | "
            << data[i].harga << " | "
            << data[i].kondisi << endl;
    }
}

void bubbleSortBrand(Sneaker data[], int jumlah){
    for(int i=0; i<jumlah-1; i++){
        for(int j=0; j<jumlah-i-1; j++){
            if(data[j].brand > data[j+1].brand){
                swap(data[j], data[j+1]);
            }
        }
    }
}

void selectionSortHarga(Sneaker data[], int jumlah){
    for(int i=0; i<jumlah-1; i++){
        int indeksMaks = i;

        for(int j=i+1; j<jumlah; j++){
            if(data[j].harga > data[indeksMaks].harga){

```

```

        indeksMaks = j;
    }
}
swap(data[i], data[indeksMaks]);
}
}

void insertionSortUkuran(Sneaker data[], int jumlah){
    for(int i=1; i<jumlah; i++){
        Sneaker kunci = data[i];
        int j = i-1;

        while(j >= 0 && data[j].detail.ukuran > kunci.detail.ukuran){
            data[j+1] = data[j];
            j--;
        }
        data[j+1] = kunci;
    }
}

// Linear Search (Brand)
int linearSearchBrand(Sneaker *data, int jumlah, string targetBrand){
    for(int indeks = 0; indeks < jumlah; indeks++){
        if(data[indeks].brand == targetBrand){
            return indeks;
        }
    }
    return -1;
}

// Binary Search (Harga)
int binarySearchHarga(Sneaker *data, int jumlah, int targetHarga){
    int indeksAwal = 0;
    int indeksAkhir = jumlah - 1;

    while(indeksAwal <= indeksAkhir){
        int indeksTengah = (indeksAwal + indeksAkhir) / 2;

        if(data[indeksTengah].harga == targetHarga){
            return indeksTengah;
        }
    }
}

```

```

    }
    else if(data[indeksTengah].harga < targetHarga){
        indeksAwal = indeksTengah + 1;
    }
    else{
        indeksAkhir = indeksTengah - 1;
    }
}
return -1;
}

```

```
// ===== MENU =====
```

```

void menu(){
    cout << "\n=== MENU ===\n";
    cout << "1. Tampil Data\n";
    cout << "2. Sorting\n";
    cout << "3. Searching\n";
    cout << "4. Keluar\n";
}

```

```
// ===== MAIN =====
```

```

int main(){

    if(!login(akun)){
        cout << "Gagal login.\n";
        return 0;
    }

    isiDataAwal(dataSneaker, jumlahData);

    int pilihan;

    do{
        menu();
        cout << "Pilih : ";
        cin >> pilihan;

        // ===== TAMPIL DATA =====
        if(pilihan == 1){
            tampilData(dataSneaker, jumlahData);
        }
    }
}

```

```

// ===== SORTING =====
else if(pilihan == 2){
    int pilihSorting;

    cout << "\n=== PILIH SORTING ===\n";
    cout << "1. Brand (Ascending - Bubble)\n";
    cout << "2. Harga (Descending - Selection)\n";
    cout << "3. Ukuran (Ascending - Insertion)\n";
    cout << "Pilih : ";
    cin >> pilihSorting;

    if(pilihSorting == 1){
        bubbleSortBrand(dataSneaker, jumlahData);
    }
    else if(pilihSorting == 2){
        selectionSortHarga(dataSneaker, jumlahData);
    }
    else if(pilihSorting == 3){
        insertionSortUkuran(dataSneaker, jumlahData);
    }

    tampilData(dataSneaker, jumlahData);
}

// ===== SEARCHING =====
else if(pilihan == 3){
    int pilihSearch;

    cout << "\n=== MENU SEARCHING ===\n";
    cout << "1. Cari Brand (Linear Search)\n";
    cout << "2. Cari Harga (Binary Search)\n";
    cout << "Pilih : ";
    cin >> pilihSearch;

    // --- Linear Search ---
    if(pilihSearch == 1){
        string cariBrand;
        cout << "Masukkan brand: ";
        cin >> cariBrand;
    }
}

```



```

        int hasil = linearSearchBrand(dataSneaker, jumlahData,
cariBrand);

        if(hasil != -1){
            cout << "Data ditemukan:\n";
            cout << dataSneaker[hasil].brand << " "
                << dataSneaker[hasil].model << endl;
        } else {
            cout << "Data tidak ditemukan\n";
        }
    }

    // --- Binary Search ---
    else if(pilihSearch == 2){
        int cariHarga;
        cout << "Masukkan harga: ";
        cin >> cariHarga;

        // WAJIB SORT DULU
        selectionSortHarga(dataSneaker, jumlahData);

        int hasil = binarySearchHarga(dataSneaker, jumlahData,
cariHarga);

        if(hasil != -1){
            cout << "Data ditemukan:\n";
            cout << dataSneaker[hasil].brand << " "
                << dataSneaker[hasil].model << endl;
        } else {
            cout << "Data tidak ditemukan\n";
        }
    }
}

}while(pilihan != 4);

return 0;
}

```

#### 4. Hasil Output

```
===== LOGIN SISTEM =====  
Username : cecil  
Password : 125  
Login berhasil!
```

Gambar 4.1 Menu Login

```
=== MENU ===  
1. Tampil Data  
2. Sorting  
3. Searching  
4. Keluar  
Pilih : █
```

Gambar 4.2 Menu Utama

```
=== Data Sneaker ===  
1. Nike AirForce1 | White 42 | 1500000 | Baru  
2. Adidas Yeezy350 | Black 43 | 4200000 | Baru  
3. Converse ChuckTaylor | Red 41 | 850000 | Bekas
```

Gambar 4.3 Tampil Data

```
=== PILIH SORTING ===  
1. Brand (Ascending - Bubble)  
2. Harga (Descending - Selection)  
3. Ukuran (Ascending - Insertion)  
Pilih : 1  
  
=== Data Sneaker ===  
1. Adidas Yeezy350 | Black 43 | 4200000 | Baru  
2. Converse ChuckTaylor | Red 41 | 850000 | Bekas  
3. Nike AirForce1 | White 42 | 1500000 | Baru
```

Gambar 4.4 Sorting Huruf (Brand - Ascending / Bubble Sort)

```
=== PILIH SORTING ===  
1. Brand (Ascending - Bubble)  
2. Harga (Descending - Selection)  
3. Ukuran (Ascending - Insertion)  
Pilih : 2  
  
=== Data Sneaker ===  
1. Adidas Yeezy350 | Black 43 | 4200000 | Baru  
2. Nike AirForce1 | White 42 | 1500000 | Baru  
3. Converse ChuckTaylor | Red 41 | 850000 | Bekas
```

Gambar 4.5 Sorting Harga (Descending / Selection Sort)

```
=== PILIH SORTING ===  
1. Brand (Ascending - Bubble)  
2. Harga (Descending - Selection)  
3. Ukuran (Ascending - Insertion)  
Pilih : 3  
  
=== Data Sneaker ===  
1. Converse ChuckTaylor | Red 41 | 850000 | Bekas  
2. Nike AirForce1 | White 42 | 1500000 | Baru  
3. Adidas Yeezy350 | Black 43 | 4200000 | Baru
```

Gambar 4.6 Sorting Ukuran (Ascending / Insertion Sort)

```
=== MENU SEARCHING ===  
1. Cari Brand (Linear Search)  
2. Cari Harga (Binary Search)  
Pilih : █
```

Gambar 4.7 Output Menu Searching

```
=== MENU SEARCHING ===  
1. Cari Brand (Linear Search)  
2. Cari Harga (Binary Search)  
Pilih : 1  
Masukkan brand: Nike  
Data ditemukan:  
Nike AirForce1
```

Gambar 4.8 Linear Search (Brand)

```
=== MENU SEARCHING ===  
1. Cari Brand (Linear Search)  
2. Cari Harga (Binary Search)  
Pilih : 2  
Masukkan harga: 1500000  
Data ditemukan:  
Nike AirForce1
```

Gambar 4.9 Binary Search (Harga)

```
=== MENU ===
1. Tampil Data
2. Sorting
3. Searching
4. Keluar
Pilih : 4
```

Gambar 4.10 Menu Keluar

## 5. Langkah-langkah GIT

### 5.1 GIT Init

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-6> git init
Initialized empty Git repository in C:/Users/MyASUS/OneDrive/文档/praktikum-apl/post-test/post-test-apl-6/.git/
```

### 5.2 GIT Add

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-6> git add .
```

### 5.3 GIT Commit

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-6> git commit -m "Posttest 6"
[master (root-commit) 1668086] Posttest 6
4 files changed, 480 insertions(+)
create mode 100644 2509106125-CeciliaMarsya-POSTTEST-6.cpp
create mode 100644 2509106125-CeciliaMarsya-POSTTEST-6.exe
```

### 5.4 GIT Push

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-6> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 27.25 KiB | 1.82 MiB/s, done.
Total 8 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
To https://github.com/cecilpua/praktikum-apl
4119d2e..92aaab7 main -> main
branch 'main' set up to track 'origin/main'.
```